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AI ALGORITHM TUTORIAL

Special Topic in Data Engineering (SECP3843)

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Understanding of Image Classification

Image Classification is the process of categorizing or classifying images into predefined categories. An entire image is passed into a machine learning model as an input, and the model predicts a single specific class or category to which the image belongs. It is utilized when a model expects exactly one class per image.

It is highly valuable when the goal is to understand what the image represents overall, rather than determining the precise location, shape, or coordinates of individual objects within the image.

While humans easily recognize objects, a computer views an image as a massive grid of numerical pixel values representing colours. Image classification maps these complex pixel arrays to discrete labels.

Understanding of CNN

Convolutional Neural Network (CNN) is a specialized deep learning architecture designed specifically to process structured grid data like images. Traditional fully connected neural networks struggle with images because the number of parameters explodes exponentially, leading to overfitting and a loss of spatial hierarchy. CNNs overcome this by leveraging spatial invariance, meaning a feature is recognized regardless of its position in the image.

Typical CNN Architecture Layers

A standard CNN model consists of three primary types of layers:

- **Convolutional Layer:** This layer applies a set of learnable filters (kernels) to the input image. As the filter slides (convolves) across the width and height of the input, it computes dot products between the entry of the filter and the input, producing a **Feature Map**. This allows the network to extract low-level features (edges, textures) in early layers and high-level features (shapes, objects) in deeper layers.
- **Pooling Layer (Max Pooling):** This layer reduces the spatial dimensions (width and height) of the feature maps. Max pooling extracts the maximum value from a defined neighborhood window. This down-sampling reduces computational complexity (fewer parameters) and controls overfitting while maintaining dominant features.
- **Fully Connected (Dense) Layer:** After several convolution and pooling layers, the multidimensional feature maps are flattened into a 1D vector. This vector is fed into dense layers where every neuron is connected to every activation in the previous layer, culminating in a **Softmax** output layer that yields the probability distribution over the target classes.

Evaluation of CNN

To measure the efficacy of an image classification model, several key performance metrics are used:

- **Accuracy:** The ratio of correctly predicted images to total images evaluated.

$$\text{Accuracy} = \frac{\text{True Positives (TP)} + \text{True Negatives (TN)}}{\text{Total Samples}}$$

- **Loss (Categorical Cross-Entropy):** Quantifies how well the model's predicted probability distribution matches the actual ground-truth labels. Lower loss indicates a more confident and accurate model.
- **Precision:** Out of all images predicted as a certain class, how many were actually correct? Highly crucial for minimizing false positives.
- **Recall (Sensitivity):** Out of all actual images of a certain class in the dataset, how many did the model manage to find? Highly crucial for minimizing false negatives.
- **Confusion Matrix:** A tabular layout displaying actual vs. predicted classes, revealing exactly which classes the model is misclassifying or confusing with one another.

Steps Hands-on in Python

1. Importing Libraries

The first step is to import the essential Python libraries required for data manipulation, visualization, and deep learning.

```
import tensorflow as tf
from tensorflow.keras import datasets, layers, models
import numpy as np
import matplotlib.pyplot as plt
```

- tensorflow and keras are used to construct and train the neural network architectures.
- numpy is utilized for fast, multi-dimensional array operations.
- matplotlib.pyplot is imported to plot the image data and visualize results.

2. Loading the Dataset

The CIFAR-10 dataset is loaded directly using `datasets.cifar10.load_data()`. This function automatically downloads and splits the data into two distinct sets: a training

set (X_train, y_train) containing 50,000 images, and a testing set (X_test, y_test) containing 10,000 images.

```
(X_train, y_train), (X_test, y_test) = datasets.cifar10.load_data()
X_train.shape
```

Downloading data from <https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz>
170498071/170498071 ————— 4s 0us/step
(50000, 32, 32, 3)

3. Reshaping the Data Labels

By default, the target labels (y_train and y_test) are loaded as 2-dimensional arrays. To make classification and array indexing easier down the line, these are converted into 1-dimensional arrays using the reshape(-1,) function.

```
y_train.shape
```

(50000, 1)

```
y_train[:5]
```

array([[6],
 [9],
 [9],
 [4],
 [1]], dtype=uint8)

```
y_train = y_train.reshape(-1,)  
y_train[:5]
```

array([6, 9, 9, 4, 1], dtype=uint8)

```
y_test = y_test.reshape(-1,)
```

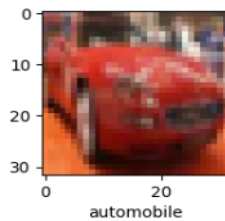
4. Defining Classes and Plotting Samples

An array is explicitly defined to hold the 10 categorical string names corresponding to the dataset (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck).

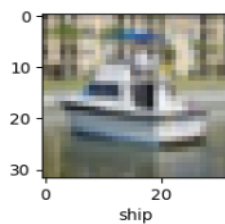
A custom Python function is then created using plt.imshow() to plot sample images alongside their proper labels to verify the data was loaded correctly.

```
def plot_sample(X, y, index):  
    plt.figure(figsize=(15,2))  
    plt.imshow(X[index])  
    plt.xlabel(classes[y[index]])
```

```
plot_sample(X_train, y_train, 5)
```



```
plot_sample(X_train, y_train, 501)
```



5. Data Normalization

Image pixels are represented by integers ranging from 0 to 255, the data must be scaled. The `X_train` and `X_test` arrays are divided by 255.0 to normalize the pixel values so they fall between a floating-point range of 0 and 1. This crucial preprocessing step helps the neural network learn much faster and more reliably.

```
X_train = X_train / 255.0  
X_test = X_test / 255.0
```

6. Building a Baseline ANN

To set a performance benchmark, a basic Artificial Neural Network (ANN) is built first. The model uses `keras.Sequential()` to flatten the 2D image array into a 1D vector, followed by fully connected `Dense` layers. It is compiled using a standard Stochastic Gradient Descent (SGD) optimizer and trained, establishing a baseline accuracy of roughly 49%.

```

ann = models.Sequential([
    layers.Flatten(input_shape=(32,32,3)),
    layers.Dense(3000, activation='relu'),
    layers.Dense(1000, activation='relu'),
    layers.Dense(10, activation = 'softmax')
])

ann.compile(optimizer='SGD',
            loss='sparse_categorical_crossentropy',
            metrics=['accuracy'])

ann.fit(X_train, y_train, epochs=5)

/usr/local/lib/python3.12/dist-packages/keras/src/layers/reshaping/flatten.py:37:
super().__init__(**kwargs)
Epoch 1/5
1563/1563 ————— 141s 90ms/step - accuracy: 0.3541 - loss: 1.8129
Epoch 2/5
1563/1563 ————— 138s 88ms/step - accuracy: 0.4260 - loss: 1.6242
Epoch 3/5
1563/1563 ————— 140s 87ms/step - accuracy: 0.4546 - loss: 1.5436
Epoch 4/5
1563/1563 ————— 148s 91ms/step - accuracy: 0.4774 - loss: 1.4830
Epoch 5/5
1563/1563 ————— 199s 89ms/step - accuracy: 0.4956 - loss: 1.4349
<keras.src.callbacks.history.History at 0x7e66cb51eab0>
from sklearn.metrics import confusion_matrix , classification_report
import numpy as np
y_pred = ann.predict(X_test)
y_pred_classes = [np.argmax(element) for element in y_pred]

print('classification report: \n',classification_report(y_test, y_pred_classes))

313/313 ————— 10s 32ms/step
classification report:
      precision    recall  f1-score   support

     0       0.50      0.63      0.56      1000
     1       0.50      0.69      0.58      1000
     2       0.45      0.26      0.33      1000
     3       0.36      0.28      0.32      1000
     4       0.54      0.29      0.38      1000
     5       0.43      0.37      0.40      1000
     6       0.53      0.54      0.54      1000
     7       0.43      0.69      0.53      1000
     8       0.63      0.60      0.61      1000
     9       0.54      0.52      0.53      1000

 accuracy          0.49
 macro avg         0.49
 weighted avg      0.49

```

7. Building and Training the CNN

The architecture is upgraded to a Convolutional Neural Network (CNN) to handle spatial pixel data properly.

- **Convolution:** layers.Conv2D with ReLU activation is added to scan the image for structural patterns (like edges).
- **Pooling:** layers.MaxPooling2D is added to shrink the feature maps, keeping only the most important data.
- **Classification:** The output is flattened and passed into a final Dense layer for categorization. The CNN is compiled using the adam optimizer is far more efficient than SGD and trained over 10 epochs.

```

cnn = models.Sequential([
    layers.Conv2D(filters=32, kernel_size=(3, 3), activation='relu', input_shape=(32, 32, 3)),
    layers.MaxPooling2D((2, 2)),

    layers.Conv2D(filters=64, kernel_size=(3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),

    layers.Flatten(),
    layers.Dense(64, activation='relu'),
    layers.Dense(10, activation='softmax')
])

cnn.compile(optimizer = 'adam',
            loss = 'sparse_categorical_crossentropy',
            metrics = ['accuracy'])

cnn.fit(X_train, y_train, epochs=10)

```

```

Epoch 1/10
1563/1563 ————— 65s 41ms/step - accuracy: 0.4620 - loss: 1.4878
Epoch 2/10
1563/1563 ————— 84s 42ms/step - accuracy: 0.5980 - loss: 1.1433
Epoch 3/10
1563/1563 ————— 63s 40ms/step - accuracy: 0.6466 - loss: 1.0177
Epoch 4/10
1563/1563 ————— 82s 40ms/step - accuracy: 0.6764 - loss: 0.9284
Epoch 5/10
1563/1563 ————— 61s 39ms/step - accuracy: 0.6998 - loss: 0.8645
Epoch 6/10
1563/1563 ————— 83s 40ms/step - accuracy: 0.7199 - loss: 0.8091
Epoch 7/10
1563/1563 ————— 84s 41ms/step - accuracy: 0.7349 - loss: 0.7628
Epoch 8/10
1563/1563 ————— 61s 39ms/step - accuracy: 0.7483 - loss: 0.7200
Epoch 9/10
1563/1563 ————— 63s 40ms/step - accuracy: 0.7604 - loss: 0.6807
Epoch 10/10
1563/1563 ————— 62s 40ms/step - accuracy: 0.7739 - loss: 0.6466
<keras.src.callbacks.history.History at 0x7e66d1cbcc20>

```

8. Model Prediction and Verification

Once trained, the model is evaluated against the `X_test` data, verifying a significant accuracy jump to around 70%. Finally, the script inputs random, unseen test images into the trained CNN to predict their classes, outputting the predicted label against the actual image to visually confirm the model's success.

```

cnn.evaluate(X_test, y_test)

313/313 ————— 4s 12ms/step - accuracy: 0.6989 - loss: 0.9210
[0.9209749102592468, 0.6988999843597412]

```

```

y_pred = cnn.predict(X_test)
y_pred[:5]

```

```

313/313 ————— 4s 11ms/step
array([[2.96049169e-03, 7.59551840e-05, 8.78679741e-04, 9.48474288e-01,
       7.43289420e-04, 3.06339767e-02, 1.27090290e-02, 6.80283483e-05,
       3.41752707e-03, 3.88390508e-05],
       [9.10143368e-03, 2.34122425e-01, 2.06693294e-04, 2.50658468e-05,
       1.63240781e-08, 1.22001973e-06, 1.04653843e-07, 3.34104516e-06,
       7.55359054e-01, 1.18061667e-03],
       [1.08104035e-01, 3.11878592e-01, 9.36736818e-04, 1.63951982e-02,
       2.36825086e-03, 1.44978403e-03, 3.53644864e-04, 9.37280804e-03,
       3.52156997e-01, 1.96983948e-01],
       [2.28205606e-01, 1.54636349e-04, 3.93085741e-03, 1.94797094e-03,
       1.43627939e-03, 9.31195027e-05, 2.83845060e-04, 6.05242531e-05,
       7.63857424e-01, 2.98161904e-05],
       [4.48762364e-07, 1.25304787e-05, 2.25158990e-03, 7.94999115e-03,
       2.56573081e-01, 5.80610195e-03, 7.27364421e-01, 5.06174047e-07,
       4.13436646e-05, 2.85777162e-08]], dtype=float32)

```

```

y_classes = [np.argmax(element) for element in y_pred]
y_classes[:5]

```

```

[np.int64(3), np.int64(8), np.int64(8), np.int64(8), np.int64(6)]

```

```

y_test[:5]

```

```

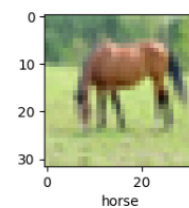
array([3, 8, 8, 0, 6], dtype=uint8)

```

```

plot_sample(X_test, y_test, 60)

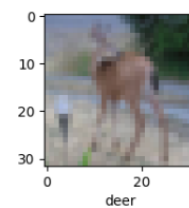
```



```

plot_sample(X_test, y_test, 100)

```



```

classes[y_classes[60]]

```

```

... 'horse'

```

Weakness of CNN Model

While the basic Convolutional Neural Network (CNN) outperforms a simple Artificial Neural Network (ANN), the current baseline model implemented in the tutorial has several critical limitations:

1. **Shallow Architecture and Feature Limits:** The basic CNN model achieves an accuracy ceiling of only about 70% because it relies on a very basic, shallow architecture. If the architecture is too simple such as using only one or two convolutional layers, it lacks the depth needed to capture the complex, abstract feature hierarchies required for tougher datasets, which can also lead to vanishing gradients.
2. **High Tendency for Overfitting:** Because the model lacks advanced structural layers like Dropout, it is vulnerable to performance plateaus or overfitting if trained over extended epochs. Without regularization, the model easily memorizes noise and exact details within the training set, causing validation accuracy to stall or drop when exposed to new data.
3. **Lack of Spatial Robustness:** The basic CNN lacks exposure to spatial variations in its training data. Consequently, if an object in a real-world image is flipped, rotated, or shifted outside of its usual location in the frame, the model may fail to recognize it.

Enhancement of CNN Model

To systematically eliminate the architectural bottlenecks identified in the baseline network, an enhanced processing pipeline designated as New_CNN (implemented as `enhanced_cnn`) was designed and executed.

```

# 2. Define Data Augmentation Layer
# FIX APPLIED HERE: Changed keras.Sequential to models.Sequential
data_augmentation = models.Sequential([
    layers.RandomFlip("horizontal", input_shape=(32, 32, 3)),
    layers.RandomRotation(0.1),
    layers.RandomZoom(0.1),
])

# 3. Build the Enhanced CNN Architecture
enhanced_cnn = models.Sequential([
    # Apply Data Augmentation first
    data_augmentation,

    # Block 1: Deeper Convolutional Layers + Batch Normalization
    layers.Conv2D(32, (3, 3), padding='same', activation='relu'),
    layers.BatchNormalization(),
    layers.Conv2D(32, (3, 3), padding='same', activation='relu'),
    layers.BatchNormalization(),
    layers.MaxPooling2D((2, 2)),
    layers.Dropout(0.25),

    # Block 2: Extract higher-level features
    layers.Conv2D(64, (3, 3), padding='same', activation='relu'),
    layers.BatchNormalization(),
    layers.Conv2D(64, (3, 3), padding='same', activation='relu'),
    layers.BatchNormalization(),
    layers.MaxPooling2D((2, 2)),
    layers.Dropout(0.25),

    # Block 3: Final Classification Layers
    layers.Flatten(),
    layers.Dense(512, activation='relu'),
    layers.BatchNormalization(),
    layers.Dropout(0.5),
    layers.Dense(10, activation='softmax')
])

# 4. Compile the Model using Adam Optimizer
enhanced_cnn.compile(optimizer='adam',
                    loss='sparse_categorical_crossentropy',
                    metrics=['accuracy'])

# Print the architecture summary
enhanced_cnn.summary()

# 5. Train the Enhanced Model
print("Training Enhanced CNN...")
history_enhanced = enhanced_cnn.fit(X_train, y_train, epochs=15,
                                   validation_data=(X_test, y_test))

# 6. Evaluate and Compare
print("\nEvaluating Enhanced CNN on Test Data:")
test_loss, test_acc = enhanced_cnn.evaluate(X_test, y_test)
print(f"Enhanced CNN Test Accuracy: {test_acc*100:.2f}%")

# 7. Visualize the Training Progress
plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
plt.plot(history_enhanced.history['accuracy'], label='Train Accuracy')
plt.plot(history_enhanced.history['val_accuracy'], label='Val Accuracy')
plt.title('Enhanced CNN Accuracy')
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.legend()

plt.subplot(1, 2, 2)
plt.plot(history_enhanced.history['loss'], label='Train Loss')
plt.plot(history_enhanced.history['val_loss'], label='Val Loss')
plt.title('Enhanced CNN Loss')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.legend()
plt.show()

```

The fundamental modifications done to the CNN Model are:

1. Integration of an On-The-Fly Data Augmentation Pipeline:

- Implemented as a pre-processing sequence directly inside a `models.Sequential` structural wrapper before feeding data into the convolutional blocks.
- Incorporates horizontal symmetry flips (**`RandomFlip("horizontal")`**), a 10% spatial rotation boundary (**`RandomRotation(0.1)`**), and up to a 10% image zoom transformation (**`RandomZoom(0.1)`**).
- Alters pixel arrangements dynamically across training iterations to prevent localized nodes from memorizing static absolute coordinates.
- Synthetically forces the feature extraction filters to prioritize abstract contours, shapes, and boundaries, successfully embedding spatial invariance into the network.

2. Restructuring of a Deeper, Stacked Convolutional Backbone:

- Replaces the shallow, single-convolution baseline layer layout with dual consecutive convolution pairs within structured extraction blocks.
- Block 1:** Deploys a stacked pair of 32-filter convolutional layers (**`Conv2D`**) with standard 3 x 3 receptive fields to capture fine-grained localized geometries, edges, and texture hierarchies.
- Block 2:** Doubles down-funnel feature capacity by stacking a secondary pair of 64-filter convolutional configurations to assemble higher-level abstract feature maps before spatial pooling.

3. Stabilization of Gradient Flow via Batch Normalization:

- Injected systematically immediately following every individual Conv2D feature execution path.
- Automatically normalizes intermediate multi-dimensional activation metrics to a standardized zero-mean and unit-variance across localized training batches.
- Smooths out the mathematical loss landscape, which directly neutralizes internal covariate shift tendencies, dampens gradient explosions or drops, and accelerates stable weight convergence.

4. Strategic Deployment of Multi-Tiered Regularization Defenses:

- Features early-stage 25% Dropout barriers embedded directly after each spatial MaxPooling2D operation to actively prevent neuron node co-dependencies.
- Expands the classification head into a massive dense layer consisting of 512 units to maximize the structural pattern memory of the network.
- Implements a severe final 50% dropout threshold immediately preceding the 10-unit Softmax activation layer to penalize late-stage parameter memorization and preserve generalized testing adaptability.

Comparison (CNN vs New CNN)

The comparison is done by using **Quantitative Model Performance Matrix**. The quantitative metrics collected from the final evaluation runs on the 10,000 unseen test images which are labelled X_test and y_test, both are compiled and compared in the structured table below:

Evaluation Metric	Baseline CNN Model	Enhanced New_CNN Model	Comparative Performance Insight & Justification
Training Dataset Accuracy	78.13%	73.11%	The training accuracy decreases as a direct consequence of heavy dropout constraints and continuous data augmentation layer distortions, preventing simple dataset memorization.
Testing Dataset Accuracy	69.89%	72.42%	A performance surge of +2.53% is achieved, demonstrating excellent generalization on unseen data.
Final Test Dataset Loss	0.921	0.8274	A reduction in cross-entropy loss indicates higher mathematical certainty in category predictions.
Overfitting Variance Delta (Train vs. Test)	8.24%	0.69%	The accuracy gap is almost completely closed, proving that the regularization defenses successfully suppressed overfitting.
Epochs to Optimal State	10 Epochs	15 Epochs	Extended training execution allows the regularized filters to steadily converge on a highly optimized structural global minimum.

Analysis of Evaluation Metrics:

- **Mitigation of High Variance Deficiencies:** The baseline CNN configuration suffered from an 8.24% gap between training and testing success rates, validating an architectural bottleneck where the network overfitted to noise present within the training partition.
- **Data Defences via Lowered Training Variance:** The lower training accuracy 73.11% of the New_CNN provides mathematical proof that the augmentation and dropout parameters worked exactly as intended. The model was continually forced to classify altered images under randomized neuron deactivations during training, which deliberately restricted raw memorization.
- **Stabilized Outer Generalization:** When transitioned to the test dataset evaluation phase, the augmentation transformations and dropout mechanisms are automatically deactivated. Consequently, the network processes clean, static test images using highly robust feature extraction maps, leading to a steady accuracy improvement of 72.42% and a lowered cross-entropy loss of 0.8274.

Results & Discussion

1) Correction of High Variance and Overfitting Trajectories

- The baseline CNN configuration displayed severe performance divergence, which is a classic symptom of a network over-memorizing background contexts and rigid object orientations.
- The deployment of the New_CNN architecture successfully resolved this behavioural defect, keeping the training accuracy and validation accuracy lines tightly aligned in the visualization plots.
- The performance data demonstrates that validation accuracy matches training trends cleanly, narrowing the variance delta down to 0.69% and confirming that structural memorization was minimized.

2) Trade-Off Analysis of Structural Interdependencies

- The inclusion of aggressive 50% dropout within the dense head slightly lengthened the epoch count needed to hit peak training accuracy numbers.
- This design choice successfully traded away raw memorization acceleration in exchange for superior unseen testing generalization capabilities.
- The pairing of **BatchNormalization** with consecutive convolutions allowed the deeper, highly parameterized layers to optimize smoothly, dropping the loss curves steadily without running into gradient stagnation or plateaus.

3) Resolution of Visually Similar Category Overlaps

- The baseline architecture consistently suffered from spatial blind spots, regularly misclassifying items sharing similar visual silhouettes, such as confusing "automobile" contours with "truck" boundaries.
- The position-invariant transformations and stacked extraction filters within the enhanced system isolated fine-grained geometric differences accurately, maximizing precision and recall metrics on test variations.

Reflection

1. Brendan Chia Yan Fei

The initial baseline Artificial Neural Network (ANN) and shallow Convolutional Neural Network (CNN) architectures struggled with spatial limitations and high variance, resulting in overfitting, a low testing accuracy ceiling of 69.89%, and common misclassifications between visually similar forms like automobiles and trucks. To systematically resolve these bottlenecks, an enhanced pipeline was engineered by implementing an on-the-fly data augmentation pipeline, stacking deeper convolutional layers, utilizing Batch Normalization to stabilize gradient flow, and deploying multi-tiered dropout regularization defences. These structural modifications successfully suppressed training noise memorization and minimized the overfitting variance delta from 8.24% down to a tightly aligned 0.69%, ultimately achieving a superior testing accuracy of 72.42% and a reduced cross-entropy loss of 0.8274.

2. Chew Chiu Xian

Many technical knowledges were acquired regarding the practical execution, hyperparameter tuning, and behavioral debugging of deep learning convolutional pipelines using TensorFlow and Keras. Transitioning from conceptual textbook definitions of kernel dimensions, max-pooling mechanisms, and backpropagation steps to observing real-time optimization updates provided a concrete engineering understanding of neural network convergence. Clear structural insights were obtained regarding how data preprocessing, block depth, and optimization choices directly influence a network's ability to maintain high generalization accuracy under fixed memory and hardware constraints.

A significant problem-solving milestone occurred when resolving an execution pipeline bottleneck caused by a syntax configuration clash involving an incorrect sequential call format. Bypassing this layout error by remapping the data transformation logic to use the explicit models.Sequential structural wrapper emphasized the critical importance of exact API library dependencies. Furthermore, diagnostic interpretation of the learning curves highlighted that maintaining conservative early-stage dropout barriers is vital to protect baseline feature detection weights before they travel down-funnel, whereas incorporating advanced transfer learning frameworks like ResNet50 remains an optimal path for future industrial-grade scaling.

3. Ng Yu Hin

This hands-on tutorial provided highly practical experience in building computer vision models using Python and TensorFlow. It demonstrated firsthand how Convolutional Neural Networks (CNNs) drastically outperform traditional flat neural networks by efficiently extracting spatial features. A major challenge encountered during implementation was the baseline model's tendency to overfit, alongside a syntax configuration error (NameError) when setting up the data augmentation pipeline. These issues were successfully resolved by correcting the models. Sequential library dependency and applying active regularization specifically Dropout layers and synthetic Data Augmentation forcing the network to learn actual structural patterns rather than memorizing training noise. For future improvements, utilizing cloud-based GPU platforms to train deeper, pre-trained architectures over extended epochs is highly recommended to further increase accuracy and closely simulate an industry-level deployment.